Uncertainty Aware Portfolio Optimisation using Machine Learning

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Assignment 1

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Question 1

(a) Simple Monthly Returns

The simple return is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Therefore,

$$R_1 = \frac{108 - 100}{100} = 0.08 = 8.00\%.$$

$$R_2 = \frac{103 - 108}{108} = -0.046296 = -4.63\%.$$

$$R_3 = \frac{115 - 103}{103} = 0.116505 = 11.65\%.$$

$$R_4 = \frac{110 - 115}{115} = -0.043478 = -4.35\%.$$

Thus, the four simple returns are

$$\boxed{(0.0800,\ -0.0463,\ 0.1165,\ -0.0435).}$$

(b) Log Returns

The log return is

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right).$$

Hence,

$$r_1 = \ln\left(\frac{108}{100}\right) = \ln(1.08) \approx 0.07696.$$

$$r_2 = \ln\left(\frac{103}{108}\right) = \ln(0.953704) \approx -0.04740.$$

$$r_3 = \ln\left(\frac{115}{103}\right) = \ln(1.116505) \approx 0.11019.$$

$$r_4 = \ln\left(\frac{110}{115}\right) = \ln(0.956522) \approx -0.04445.$$

Therefore,

$$\boxed{(0.07696,\ -0.04740,\ 0.11019,\ -0.04445).}$$

(c) Comparing r_t and R_t

The approximation

$$r_t \approx R_t$$

is expected to hold when returns are small.
The absolute errors are

$$|R_1 - r_1| = |0.08000 - 0.07696| = 0.00304,$$

$$|R_2 - r_2| = |-0.04630 - (-0.04740)| = 0.00110,$$

$$|R_3 - r_3| = |0.11651 - 0.11019| = 0.00632,$$

$$|R_4 - r_4| = |-0.04348 - (-0.04445)| = 0.00097.$$

These are summarized below:

Month	R_t	r_t	$\lvert R_t - r_t\rvert$
1	0.08000	0.07696	0.00304
2	−0.04630	−0.04740	0.00110
3	0.11651	0.11019	0.00632
4	−0.04348	−0.04445	0.00097

The approximation is *least accurate in Month 3*, where the return is largest in magnitude:

$$R_3 \approx 11.65\%.$$

This occurs because the approximation

$$\ln(1 + R) = R + O(R)$$

comes from the first-order Taylor expansion of the logarithm and becomes less accurate as R increases as $O(R)$ increases.

Question 2

(a) Expected Portfolio Return

The linearity of expectation states that for random variables X and Y and constants a and b,

$$E[aX + bY] = aE[X] + bE[Y].$$

Applying this property to the portfolio return,

$$E[R_p] = E[w_1R_1 + w_2R_2] = w_1E[R_1] + w_2E[R_2].$$

Substituting the given values,

$$E[R_p] = 0.4(0.15) + 0.6(0.07) = 0.06 + 0.042 = 0.102.$$

Therefore,

$$\boxed{E[R_p] = 0.102 = 10.2\%.}$$

(b) Effect of Correlation

The result does *not* change if R_1 and R_2 are correlated.

This is because the linearity of expectation holds regardless of whether the random variables are independent or correlated.

Hence,

$$E[R_p] = w_1E[R_1] + w_2E[R_2] = 10.2\%$$

for any correlation between R_1 and R_2.

Question 3

Let the portfolio consist of two assets with weights w_1 and w_2. The portfolio return is

$$R_p = w_1R_1 + w_2R_2.$$

We begin with the variance identity

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

Let

$$X = w_1R_1, \qquad Y = w_2R_2.$$

Then

$$\mathrm{Var}(R_p) = \mathrm{Var}(w_1R_1 + w_2R_2).$$

Applying the variance identity,

$$\mathrm{Var}(R_p) = \mathrm{Var}(w_1R_1) + \mathrm{Var}(w_2R_2) + 2\,\mathrm{Cov}(w_1R_1, w_2R_2).$$

Using the properties

$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$$

and

$$\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y),$$

we obtain

$$\mathrm{Var}(R_p) = w_1^2\,\mathrm{Var}(R_1) + w_2^2\,\mathrm{Var}(R_2) + 2w_1w_2\,\mathrm{Cov}(R_1, R_2).$$

Since

$$\mathrm{Var}(R_1) = \sigma_1^2, \qquad \mathrm{Var}(R_2) = \sigma_2^2,$$

we have

$$\mathrm{Var}(R_p) = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\mathrm{Cov}(R_1, R_2).$$

Using the relationship between covariance and correlation,

$$\mathrm{Cov}(R_1, R_2) = \rho_{12}\sigma_1\sigma_2,$$

substituting into the above expression gives

$$\mathrm{Var}(R_p) = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2.$$

Therefore, the two-asset portfolio variance formula is

$$\boxed{\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2.}$$

Question 4

Given

$$\mu_A = 18\%, \qquad \sigma_A = 25\% = 0.25,$$

$$\mu_B = 8\%, \qquad \sigma_B = 12\% = 0.12,$$

and portfolio weights

$$w_A = 0.5, \qquad w_B = 0.5.$$

The two-asset portfolio variance formula is

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\rho\sigma_A\sigma_B.$$

Substituting the given values,

$$\sigma_p^2 = (0.5)^2(0.25)^2 + (0.5)^2(0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12).$$

Since

$$(0.5)^2(0.25)^2 = 0.015625,$$

$$(0.5)^2(0.12)^2 = 0.0036,$$

$$2(0.5)(0.5)(0.25)(0.12) = 0.015,$$

we obtain

$$\sigma_p^2 = 0.019225 + 0.015\rho.$$

(a) Portfolio Risk for Different Correlations

Case 1: $\rho = +1$

$$\sigma_p^2 = 0.019225 + 0.015 = 0.034225.$$

$$\sigma_p = \sqrt{0.034225} = 0.185 = 18.5\%.$$

Case 2: $\rho = 0$

$$\sigma_p^2 = 0.019225.$$

$$\sigma_p = \sqrt{0.019225} \approx 0.13865 = 13.87\%.$$

Case 3: $\rho = -1$

$$\sigma_p^2 = 0.019225 - 0.015 = 0.004225.$$

$$\sigma_p = \sqrt{0.004225} = 0.065 = 6.5\%.$$

Hence,

$$\boxed{\begin{aligned} \rho = +1 &: \sigma_p = 18.5\%, \\ \rho = 0 &: \sigma_p \approx 13.87\%, \\ \rho = -1 &: \sigma_p = 6.5\%. \end{aligned}}$$

(b) When Does σ_p Equal the Weighted Average of Risks?

The weighted average of the individual risks is

$$w_A\sigma_A + w_B\sigma_B.$$

Consider the portfolio variance formula with $\rho = 1$:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B.$$

Notice that

$$w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B = (w_A\sigma_A + w_B\sigma_B)^2.$$

Therefore,

$$\sigma_p = \sqrt{(w_A\sigma_A + w_B\sigma_B)^2}.$$

Since standard deviations and weights are non-negative,

$$\boxed{\sigma_p = w_A\sigma_A + w_B\sigma_B}$$

when

$$\boxed{\rho = 1.}$$

Thus, portfolio risk equals the weighted average of individual risks only when the assets are perfectly positively correlated.

(c) Weight that Makes Portfolio Risk Zero for $\rho = -1$

For perfectly negatively correlated assets,

$$\sigma_p = |w_A\sigma_A - w_B\sigma_B|.$$

To obtain zero portfolio risk, we require

$$w_A\sigma_A - w_B\sigma_B = 0.$$

Since

$$w_B = 1 - w_A,$$

we get

$$w_A\sigma_A = (1 - w_A)\sigma_B.$$

Substituting the values,

$$0.25w_A = 0.12(1 - w_A).$$

Expanding,

$$0.25w_A = 0.12 - 0.12w_A.$$

$$0.37w_A = 0.12.$$

$$w_A = \frac{0.12}{0.37} = 0.3243.$$

Therefore,

$$\boxed{w_A^* = \frac{\sigma_B}{\sigma_A + \sigma_B} = \frac{0.12}{0.25 + 0.12} = 0.3243}$$

or approximately

$$\boxed{w_A^* = 32.43\%.}$$

The corresponding weight in Asset B is

$$\boxed{w_B^* = 1 - w_A^* = 67.57\%.}$$

Question 5

When two assets have correlation ($\rho < 1$), they do not move perfectly together. Looking at the portfolio variance formula,

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2,$$

The only term involving correlation is the covariance term ($2w_1w_2\rho\sigma_1\sigma_2$). If ($\rho = 1$), this term is as large as possible and the portfolio risk equals the weighted average of the individual risks. When ($\rho < 1$), the covariance term becomes smaller, reducing the overall portfolio variance. Intuitively, the assets do not rise and fall in perfect sync, so some of the fluctuations of one asset are offset by the other. This diversification effect lowers portfolio risk without necessarily lowering expected return. The smaller the correlation, the greater the risk reduction, with the strongest diversification occurring when ($\rho = -1$).

Question 6

The given asset parameters are

Asset	μ_i	σ_i
1	15%	$25\% = 0.25$
2	8%	$12\% = 0.12$
3	5%	$4\% = 0.04$

and the correlations are

$$\rho_{12} = 0.4, \qquad \rho_{13} = 0.1, \qquad \rho_{23} = 0.2,$$

with

$$\rho_{11} = \rho_{22} = \rho_{33} = 1.$$

(a) Construction of the Covariance Matrix

Recall that

$$\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j.$$

First, compute the diagonal entries:

$$\Sigma_{11} = \sigma_1^2 = (0.25)^2 = 0.0625,$$

$$\Sigma_{22} = \sigma_2^2 = (0.12)^2 = 0.0144,$$

$$\Sigma_{33} = \sigma_3^2 = (0.04)^2 = 0.0016.$$

Next, compute the off-diagonal entries:

$$\Sigma_{12} = \rho_{12}\sigma_1\sigma_2 = (0.4)(0.25)(0.12) = 0.012,$$

$$\Sigma_{13} = \rho_{13}\sigma_1\sigma_3 = (0.1)(0.25)(0.04) = 0.001,$$

$$\Sigma_{23} = \rho_{23}\sigma_2\sigma_3 = (0.2)(0.12)(0.04) = 0.00096.$$

Since covariance is symmetric,

$$\Sigma_{21} = \Sigma_{12} = 0.012,$$

$$\Sigma_{31} = \Sigma_{13} = 0.001,$$

$$\Sigma_{32} = \Sigma_{23} = 0.00096.$$

Therefore, the covariance matrix is

$$\boxed{\Sigma = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}}.$$

(b) Symmetry of Σ

A matrix is symmetric if

$$\Sigma_{ij} = \Sigma_{ji}$$

for all i, j.
From the computed entries,

$$\Sigma_{12} = \Sigma_{21} = 0.012,$$

$$\Sigma_{13} = \Sigma_{31} = 0.001,$$

$$\Sigma_{23} = \Sigma_{32} = 0.00096.$$

Hence,

$$\Sigma^T = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix} = \Sigma.$$

Therefore,

$$\boxed{\Sigma^T = \Sigma,}$$

so the covariance matrix is symmetric.

Question 7

The risk-free rate is

$$R_f = 4\% = 0.04.$$

The Sharpe Ratio is defined as

$$\text{Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}.$$

(a) Computing the Sharpe Ratios

Portfolio X

$$\text{SR}_X = \frac{0.14 - 0.04}{0.22} = \frac{0.10}{0.22} = 0.4545.$$

$$\boxed{\text{SR}_X \approx 0.455}$$

Portfolio Y

$$\text{SR}_Y = \frac{0.09 - 0.04}{0.10} = \frac{0.05}{0.10} = 0.50.$$

$$\boxed{\text{SR}_Y = 0.50}$$

Portfolio Z

$$\text{SR}_Z = \frac{0.20 - 0.04}{0.35} = \frac{0.16}{0.35} = 0.4571.$$

$$\boxed{\text{SR}_Z \approx 0.457}$$

(b) Ranking the Portfolios

The Sharpe Ratios are

$$\text{SR}_Y = 0.500, \qquad \text{SR}_Z = 0.457, \qquad \text{SR}_X = 0.455.$$

Therefore, the ranking is

$$\boxed{Y \ > \ Z \ > \ X.}$$

The portfolio with the highest Sharpe Ratio is preferred because it provides the greatest excess return per unit of risk.

Since Portfolio Y has the highest Sharpe Ratio, it offers the best risk-adjusted performance and would therefore be preferred.

(c) Why Might an Investor Prefer Y to Z?

Although Portfolio Z has the highest expected return,

$$E[R_Z] = 20\%,$$

it also carries a substantially higher risk,

$$\sigma_Z = 35\%.$$

Portfolio Y has a lower expected return,

$$E[R_Y] = 9\%,$$

but achieves a higher Sharpe Ratio:

$$0.50 > 0.457.$$

This means that Portfolio Y generates more excess return per unit of risk taken.

A rational mean-variance investor does not focus solely on return; they consider both return and risk. Therefore, despite its lower absolute return, Portfolio Y may be preferred because it offers superior risk-adjusted performance.

Question 8

(a) Global Minimum Variance (GMV) Portfolio

The Global Minimum Variance (GMV) portfolio is the portfolio with the smallest possible variance among all feasible portfolios of risky assets.

Its objective is

$$\min_{w} \; w^T \Sigma w$$

subject to the budget constraint

$$\sum_i w_i = 1.$$

The GMV portfolio is the leftmost point on the minimum variance frontier because the horizontal axis measures portfolio risk (standard deviation). Since the GMV portfolio has the lowest achievable variance, no feasible portfolio can lie further to the left. Therefore, it represents the minimum-risk portfolio among all possible combinations of the risky assets.

(b) Should a Very Risk-Averse Investor Hold a Portfolio on the Lower Half of the Frontier?

No. This statement is incorrect.

The lower half of the minimum variance frontier is inefficient because every portfolio on the lower branch is dominated by another portfolio on the upper branch with the same level of risk but a higher expected return.

Therefore, for any portfolio on the lower half, an investor can move to the corresponding portfolio on the upper half and obtain a greater expected return without increasing risk.

A very risk-averse investor may choose a portfolio close to the GMV portfolio, but it should still lie on the efficient frontier (the upper half of the minimum variance frontier), not on the lower half.

(c) Capital Market Line (CML)

The Capital Market Line (CML) is constructed by introducing a risk-free asset with return R_f and drawing a line from the point $(0, R_f)$ that is tangent to the efficient frontier of risky assets.

The slope of the CML is

$$\frac{E[R_M] - R_f}{\sigma_M},$$

which is the Sharpe Ratio of the market portfolio. It measures the excess expected return earned per unit of risk and therefore represents the reward-to-risk trade-off available in the market.

Question 9

(a) Standard Error of the Sample Mean

The standard error of the sample mean is given by

$$SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}},$$

where

$$\sigma_i = 30\% = 0.30$$

and

$$n = 36.$$

Substituting the values,

$$SE(\hat{\mu}_i) = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05.$$

Therefore,

$$\boxed{SE(\hat{\mu}_i) = 0.05 = 5\%.}$$

This standard error is large relative to typical expected stock returns. As a result, the estimated mean return is highly uncertain and can be strongly influenced by random fluctuations in the historical sample. This implies that relying solely on historical averages may lead to poor estimates of future expected returns.

(b) Error Maximisation in Mean-Variance Optimisation

Mean-Variance Optimisation is highly sensitive to estimation errors in expected returns.

The optimiser allocates capital based on the estimated values of $\hat{\mu}$. If one asset's expected return is slightly overestimated due to sampling noise, the optimiser may assign an excessively large weight to that asset. Similarly, assets with slightly underestimated returns may receive very small weights.

As a result, MVO tends to "maximise" estimation errors by turning small mistakes in the inputs into large differences in portfolio weights. This phenomenon is known as the *error maximisation* property.

Machine learning models attempt to predict future returns using a larger set of features rather than relying only on historical averages. If the predictions contain useful information about future returns, they can produce more stable and informative estimates of expected returns. Consequently, the optimiser is less likely to concentrate heavily on assets that merely appear attractive because of random noise in the historical data.

(c) Number of Parameters in the Covariance Matrix

For $N = 50$ assets, the covariance matrix is a

$$50 \times 50$$

symmetric matrix.

The number of unique parameters in a symmetric matrix is

$$\frac{N(N+1)}{2}.$$

Substituting $N = 50$,

$$\frac{50(51)}{2} = 1275.$$

Therefore,

$$\boxed{1275}$$

unique parameters must be estimated.

This creates a significant estimation challenge because a large number of parameters must be inferred from a limited amount of historical data. Estimation errors in the covariance matrix can accumulate and lead to unstable portfolio weights. As the number of assets grows, the amount of data required to estimate the covariance matrix accurately increases rapidly, making reliable estimation difficult in practice. "'